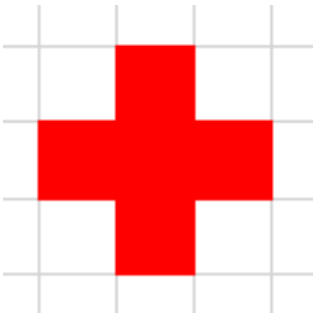
B. Fox and Cross

time limit per test 1 second
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1100 | Tags: greedy, implementation | https://codeforces.com/problemset/problem/389/B

Fox Ciel has a board with *n* rows and *n* columns. So, the board consists of *n* × *n* cells. Each cell contains either a symbol '.', or a symbol '#'.

A cross on the board is a connected set of exactly five cells of the board that looks like a cross. The picture below shows how it looks.



Ciel wants to draw several (may be zero) crosses on the board. Each cross must cover exactly five cells with symbols '#', and any cell with symbol '#' must belong to some cross. No two crosses can share a cell.

Please, tell Ciel if she can draw the crosses in the described way.

Input

The first line contains an integer *n* (3 ≤ *n* ≤ 100) — the size of the board.

Each of the next *n* lines describes one row of the board. The *i*-th line describes the *i*-th row of the board and consists of *n* characters. Each character is either a symbol '.', or a symbol '#'.

Output

Output a single line with "YES" if Ciel can draw the crosses in the described way. Otherwise output a single line with "NO".

Examples

Input
5 .#... ####. .#### ...#. .....

Output
YES

Input

```
4
####
####
####
####
```

Output

```
NO
```

Input

```
6
.#....
####..
.####.
.#.##.
######
.#..#.
```

Output

```
YES
```

Input

```
6
.#..#.
######
.####.
.####.
######
.#..#.
```

Output

```
NO
```

Input

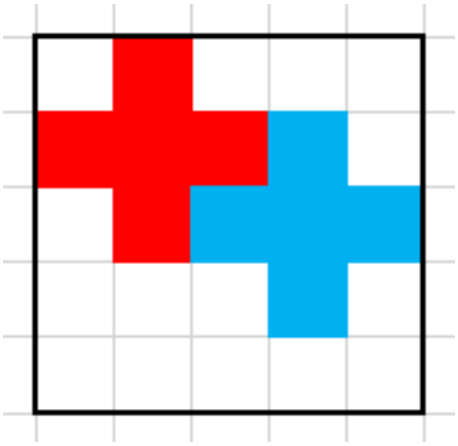
```
3
...
...
...
```

Output

```
YES
```

Note

In example 1, you can draw two crosses. The picture below shows what they look like.



In example 2, the board contains 16 cells with '#', but each cross contains 5. Since 16 is not a multiple of 5, so it's impossible to cover all.